



Spam Email Detection

SpPamm? Nahhh h

✉ Gloria Grace

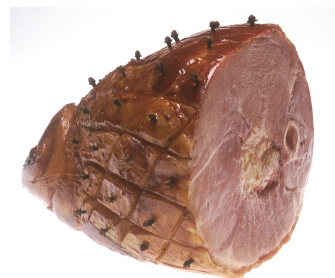
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Project Goals

- ✉ To detect and classify emails as spam or ham
- ✉ To optimize the performance of the spam email detection model by experimenting with different text preprocessing techniques



Approach

- ✘ Classification
 - ✘ Hyperparameter tuning
 - ✘ Validation approach
 - ✘ Tokenization
-
- ✘ Dataset: Enron email dataset
 - ✘ Output: a probability that gets classified as 0 or 1 (ham or spam)



The background of the image is a repeating pattern of hand-drawn icons. These icons include envelopes, some of which are open to show a letter inside, and speech bubbles containing three dots. Some envelopes are also marked with a small star. The icons are scattered across the entire image, creating a textured, patterned effect.

“DEMO”

```
# Define the new texts for prediction
new_texts = ["Hello R,\nI hope you are having a good start of your week. I wanted to address your concerns noted in this email. Firstly, from your wording of \"bed in hallway\",

# Convert the new texts into sequences of words
new_sequences = tokenizer.texts_to_sequences(new_texts)

# Pad the sequences to have the same length as the training data
new_sequences_padded = pad_sequences(new_sequences, maxlen=max_length)

# Make predictions on the new data
new_predictions = model.predict(new_sequences_padded)

# Convert the predictions to binary labels (0 or 1)
new_binary_predictions = (new_predictions > 0.5).astype(int)

# Print the predictions for the new data
for text, prediction in zip(new_texts, new_binary_predictions):
    print("Text:", text)
    if prediction == 0:
        print("Prediction: Ham")
    else:
        print("Prediction: Spam")
    print("Prob: ", new_predictions)
```

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```

1/1 [=====] - 0s 29ms/step

Text: Hello R,

I hope you are having a good start of your week. I wanted to address your concerns noted in this email. Firstly, from your wording of "bed in hallway", I don't think you are picturing

To counter your claim that there is no direct access to a window, this is frankly quite false. K sleeps with her bedroom door open, and since the closet space (my room) does not have
In fact, my room has direct access to not only one but two windows. Again, because there is not a separate door, the space that I am referring to as my room is not an enclosed space,
When G, K, and I moved in, we were told that our apartment layout (with the smaller kitchen and the walk-in closet) was due to Calvin's previous housing policies, which (we were told)
I lived off-campus in the Project Neighborhood program for the entirety of last school year, and I experienced much greater freedoms there; returning to campus has been restrictive and

All good things

A.M.

By the way, our entire apartment has been ridiculously hot since the AC was turned off last semester. Why don't you spend your time addressing actual problems that affect your residen

Prediction: Spam

Prob: [[0.6911007]]



```
# Choose one of the misclassified indices
misclassified_index = misclassified_indices[0]

# Retrieve the text of the misclassified email using the index
misclassified_email_text = X_test.iloc[misclassified_index]

# Display the contents of the misclassified email
print("Misclassified Email Text:")
print(misclassified_email_text)
print("Predicted label: ", test_binary_predictions[misclassified_indices[0]])
print("True label:", y_test_encoded[misclassified_indices[0]])
print("Prob:", y_pred_probabilities[misclassified_indices[0]])
```

Misclassified Email Text:

Subject: hunter ' s birthday party pictures

if the above link doesn ' t work , try going to [http : / / www . abacustech . net / pic - ram](http://www.abacustech.net/pic-ram) .

Predicted label: [1]

True label: 0

Prob: [0.6047787]

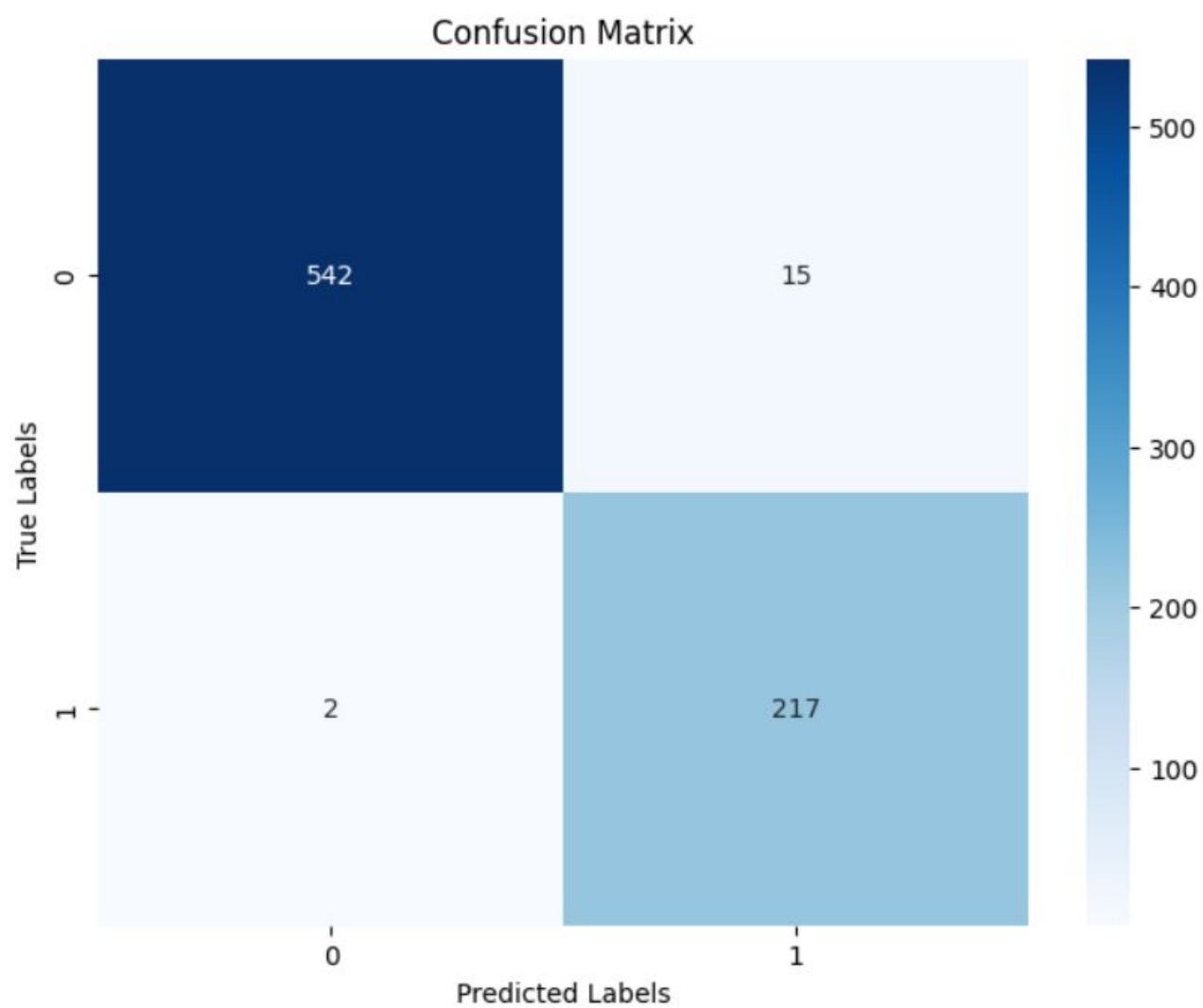
Results

The model was trained for 10 epochs, with a final accuracy of 98% on the test data.

```
Epoch 1/10
114/114 - 3s - loss: 0.5754 - accuracy: 0.6955 - val_loss: 0.4241 - val_accuracy: 0.7448 - 3s/epoch - 27ms/step
Epoch 2/10
114/114 - 2s - loss: 0.2868 - accuracy: 0.8986 - val_loss: 0.1538 - val_accuracy: 0.9704 - 2s/epoch - 19ms/step
Epoch 3/10
114/114 - 2s - loss: 0.1213 - accuracy: 0.9724 - val_loss: 0.0956 - val_accuracy: 0.9755 - 2s/epoch - 20ms/step
Epoch 4/10
114/114 - 2s - loss: 0.0771 - accuracy: 0.9776 - val_loss: 0.0775 - val_accuracy: 0.9832 - 2s/epoch - 19ms/step
Epoch 5/10
114/114 - 4s - loss: 0.0558 - accuracy: 0.9862 - val_loss: 0.0626 - val_accuracy: 0.9832 - 4s/epoch - 38ms/step
Epoch 6/10
114/114 - 2s - loss: 0.0471 - accuracy: 0.9845 - val_loss: 0.0565 - val_accuracy: 0.9794 - 2s/epoch - 19ms/step
Epoch 7/10
114/114 - 2s - loss: 0.0364 - accuracy: 0.9903 - val_loss: 0.0515 - val_accuracy: 0.9807 - 2s/epoch - 22ms/step
Epoch 8/10
114/114 - 3s - loss: 0.0311 - accuracy: 0.9917 - val_loss: 0.0558 - val_accuracy: 0.9755 - 3s/epoch - 29ms/step
Epoch 9/10
114/114 - 3s - loss: 0.0271 - accuracy: 0.9920 - val_loss: 0.0547 - val_accuracy: 0.9820 - 3s/epoch - 26ms/step
Epoch 10/10
114/114 - 3s - loss: 0.0205 - accuracy: 0.9950 - val_loss: 0.0492 - val_accuracy: 0.9858 - 3s/epoch - 27ms/step
25/25 [=====] - 0s 3ms/step - loss: 0.0486 - accuracy: 0.9781
Test loss: 0.04856019839644432
Test Accuracy: 0.9780927896499634
```



Results



What We Learned

- ✉ The distribution between the ham and spam emails.
- ✉ Machine learning is associated with various tasks such as classification, regression, clustering, and pattern recognition, among others.
- ✉ Techniques such as dropout regularization and optimization algorithms like Adam have been tried to enhance model performance.





Other Takeaways

Limitations

- ☒ Relies on the patterns of the training dataset 🚂
- ☒ Not is regularly updated
- ☒ Imbalance dataset leads to possible bias

Broader questions this project might raise:

- ☒ Privacy concerns of whether spam email detection systems analyze the content of emails is stored or shared with third parties

Future Directions:

- ☒ Collect more spam emails
- ☒ Adjusting model hyperparameters
- ☒ Explore other alternatives

